

# **An Introduction to Post-Quantum Cryptography**

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Manipal School of Information Sciences  
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Friday, 24 April 2026

- ✓ Classical Cryptography
  - Threats from Quantum Computing
  - Mitigating the Quantum Threats
- ✓ Post-Quantum Cryptography (PQC)

✓ *involves hands-on Python programming*

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|----------------------------------|--------------------|
| ✓ Classical Cryptography         | <b>~2 hours</b>    |
| – Threats from Quantum Computing | <b>~20 minutes</b> |
| – Mitigating the Quantum Threats | <b>~20 minutes</b> |
| ✓ Post-Quantum Cryptography      | <b>~1.5 hours</b>  |

✓ *involves hands-on Python programming*

- Classical Cryptography **~2 hours**
  - Primitives
  - Protocols
  - Computational Hardness Assumptions
- Threats from Quantum Computing **~20 minutes**
- Mitigating the Quantum Threats **~20 minutes**
- Post-Quantum Cryptography **~1.5 hours**
  - Quantum-Resistant Computational Problems
  - Principles of Lattice-Based Cryptography
  - ML-KEM and ML-DSA schemes

**Cryptography is fun because it is a blend of**

- Mathematics
- Computational Complexity Theory
- Algorithms and Programming
- Systems Security

## Science and engineering in equal parts

Mathematics

Computational Complexity Theory

Algorithms and Programming

Systems Security

**Do not hesitate to ask questions!**

**Why should I care?**

## Cryptography is pervasive

- Aadhaar
- UPI
- TLS (HTTPS)
- WhatsApp/Signal
- VPNs
- SSH
- GitHub
- Cryptocurrencies

## PQC Readiness In the Works...

NITI Aayog's Roadmap.

[niti\\_roadmap\\_quantum\\_powered\\_economy.pdf](#)

DST's Report

[dst\\_quantum\\_safe\\_ecosystem.pdf](#)

Society for Electronic Transactions and Security.

[pki\\_pqc\\_india.pdf](#)

CERT-In and SISA Whitepaper.

[certin\\_quantum\\_readiness.pdf](#)

UIDAI Enrollment Data Security.

Let's develop an intuitive understanding of a bunch of terms that we will be using repeatedly.

These terms are not formal definitions, but they will help us understand concepts better.

We will use simple math ideas and notation.

Don't worry if you don't understand all the terms.

Classical Computers and Algorithms

Quantum Computers and Quantum Algorithms

Post-Quantum Cryptography (PQC)

- It is **NOT post-quantum computing**
- It is *simply* post-quantum cryptography.

## Primitives and Constructions

### – Primitives

- Pseudorandom Generators (PRGs)
- Hash functions
- Symmetric-key encryption methods

### – Constructions

- Key Derivation Functions (KDFs)
- Message Authentication Codes (MACs)

## Schemes and Protocols

- Schemes
  - Public-key encryption
  - Digital signatures
- Protocols
  - Diffie-Hellman Key Exchange
  - Transport Layer Security (TLS)
  - Secure Shell (SSH)
  - Signal Messaging Protocol

# Questions?

Next Session: **Hands-on Classical Cryptography**